Pediatric Neurology II

Principles and Practices of Osteopathic Medicine 1

Thomas Chan D.O.
Associate Professor/ Dept of Clinical Specialties

tchan02@nyit.edu

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

- Discussion of hypotonia and other neurologic findings in infants and children- presentation and initial management
- Discuss various conditions with hypotonia as a primary feature and their clinical features and presentation
- Understand clinical significance of various conditions causing abnormal head shapes in infants and children
- Discuss clinical aspects related to infection
- Discuss neurologic issues associated with trauma

Source: Course Syllabus

Neurologic findings- Motor tone in Infants

- Rigidity
- Fasciculations (twitching-esp tongue)
- **Hypotonia**
- Asymmetry
- Spasticity
- Opisthotonos (severe hyperextension of the spine/ posturing)

Hypotonia ("floppy baby")

Causes of Hypotonia

- Down’s Syndrome
- Neonatal hypoxic injury
- Cerebral Palsy
- Prader-Willi syndrome
- Spinal Muscular Atrophy
- Botulism
- Myastenia Gravis
- Juvenile Dermatomyositis
- Duchenne Muscular Dystrophy
- Becker Muscular Dysthrophy
- Many other causes…

Cerebral Palsy

- **Non-progressive** motor impairment due to damage/ dysfunction of the brain or spinal tracts
- Different types- Spastic, ataxic, atheoid, mixed
- **Spastic** most common- **upper motor neuron lesion- increased deep tendon reflexes, spasticity, sustained clonus, positive babinski**
- Birth history/ prematurity/ hypoxic event
- **Hypotonic at birth and newborn period** and progressively hypertonic and hyper-reflexia

Spinal Muscular Atrophy

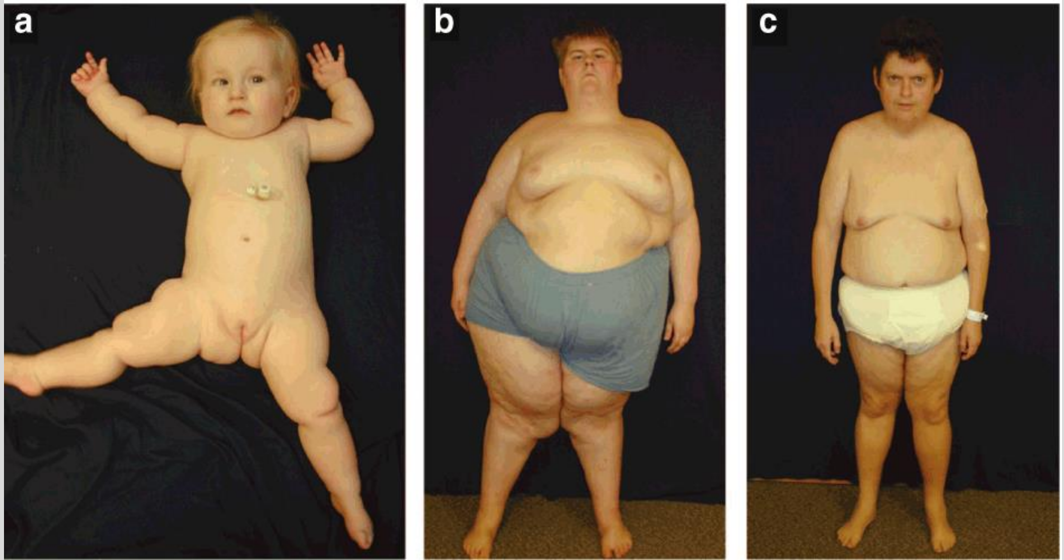
- Type **1:Wednig-Hoffman Disease-** severe infantile form
- Type 2: late infant and slowly progressive
- Type 3
- Type 0- fatal perinatal
- **Severe hypotonia**
- Thin muscle mass
- Absent DTR
- Resp distress, poor feeding
- **Tongue Fasciculations**
- Milder- wheelchair, severely handicapped
- **Normal Intelligence**

Down's Syndrome

- Common physical findings: flattened occiput, epicanthal folds, midface hypoplasia, upslanting palebral fissures, single palmar crease, widened gap between big and second toe, enlarged tongue
- Higher incidence of congenital heart disease, anomalies of GI tract (duodenal atresia), leukemia, visual and hearing impairments
- **Generalized hypotonia**
- Joint laxity

Prader-Willi Syndrome

- Deletion of paternal Chromosome 15q11.2-q13)
- Common findings: mild-moderate intellectual impairment, extreme hyperphagia, obesity, short stature, hypogonadism, poor sucking, small hands and feet
- **Infants with hypotonia and Failure to thrive, as child becomes hyperphagia and obesity** and all related complications- need extreme diet measures to control weight



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Dermatomyositis

- Common first presenting sign: violet colored rash on face (Heliotrope rash) , eyelids, and above joints (grotton’s Papules)
- Proximal muscle weakness (neck, shoulders, hips)
  - Trouble brushing hair, lifting over shoulders, climbing stairs

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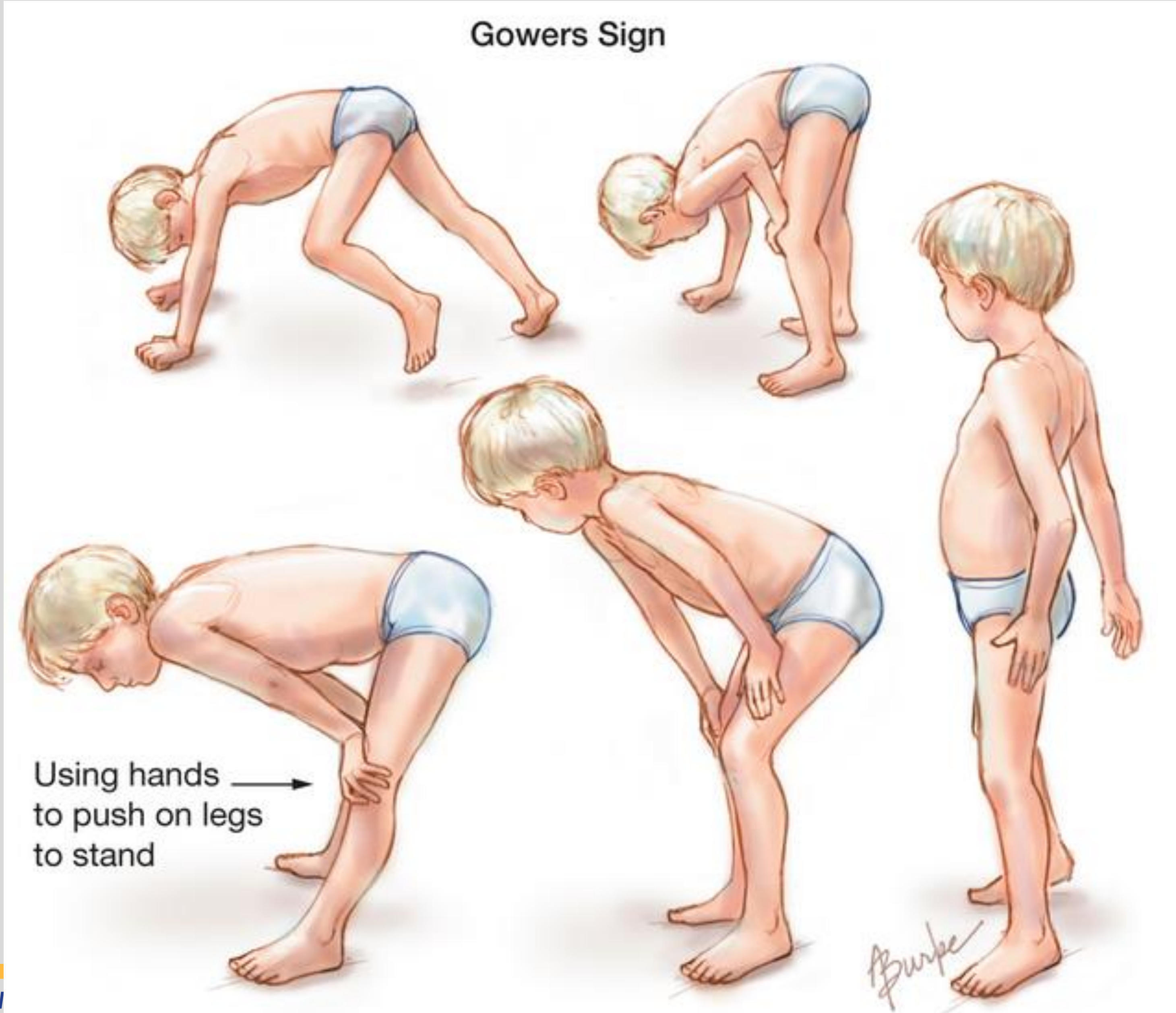
Duchenne Muscular Dystrophy Pathophysiology

- X linked recessive: frameshift mutation of dystrophin gene
- Dystrophin: subsarcolemmal protein that attaches the muscle cytoskeleton with the extracellular membrane. Works to strengthen muscle fibers
- Mutations in the dystrophin gene lead to progressive muscle fiber degeneration and weakness
- **Causes pseudohypertrophy (typically calf) by muscle breakdown, infiltration of fat, and proliferation of collagen (fibrous tissue replaces muscle)**

Duchenne Muscular Dystrophy

- Progressive proximal weakness, intellectual impairment, calf pseudohypertrophy, wasting of thigh muscles, cardiomyopathy
- Early signs: poor head control in infancy, hip girdle weakness around age 2, Gower sign
- **Significant elevation of serum Creatine Kinase(CK)**
- Death depends on severity and care but significantly shortened lifespan (respiratory failure, heart failure, pneumonia, aspiration).

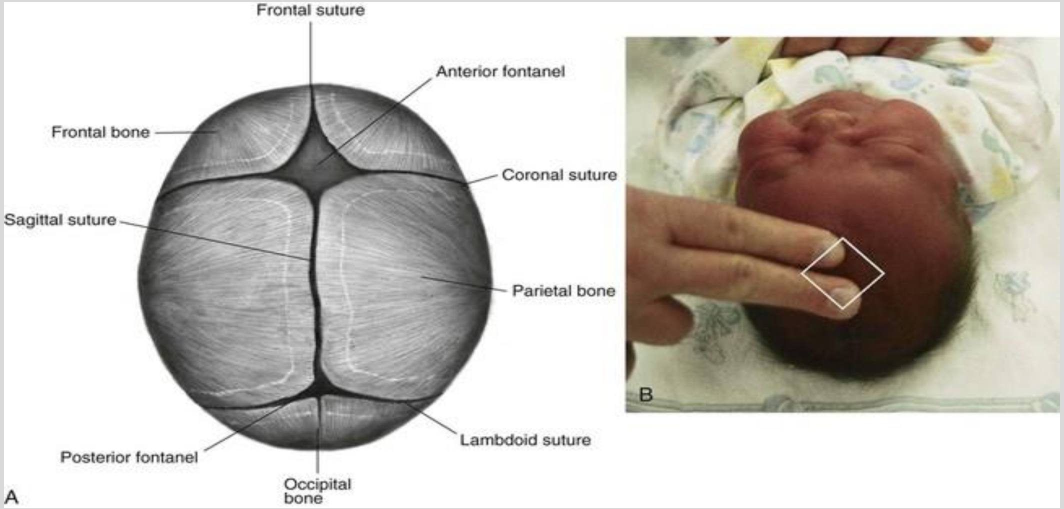

Gowers Sign
Using hands
to push on legs
to stand

Becker Muscular Dystrophy

- X linked recessive. Same genetic defect as Duchenne Muscular Dystrophy, except some functional dystrophin is produced → less severe overall
- Ambulatory until late adolescence
- Similar symptoms as Duchenne Muscular Dystrophy

Baby Head…What's normal

- Anterior Fontanel (6-18months close)
- Posterior Fontanel (6-8weeks close)
- Sutures
- Molding



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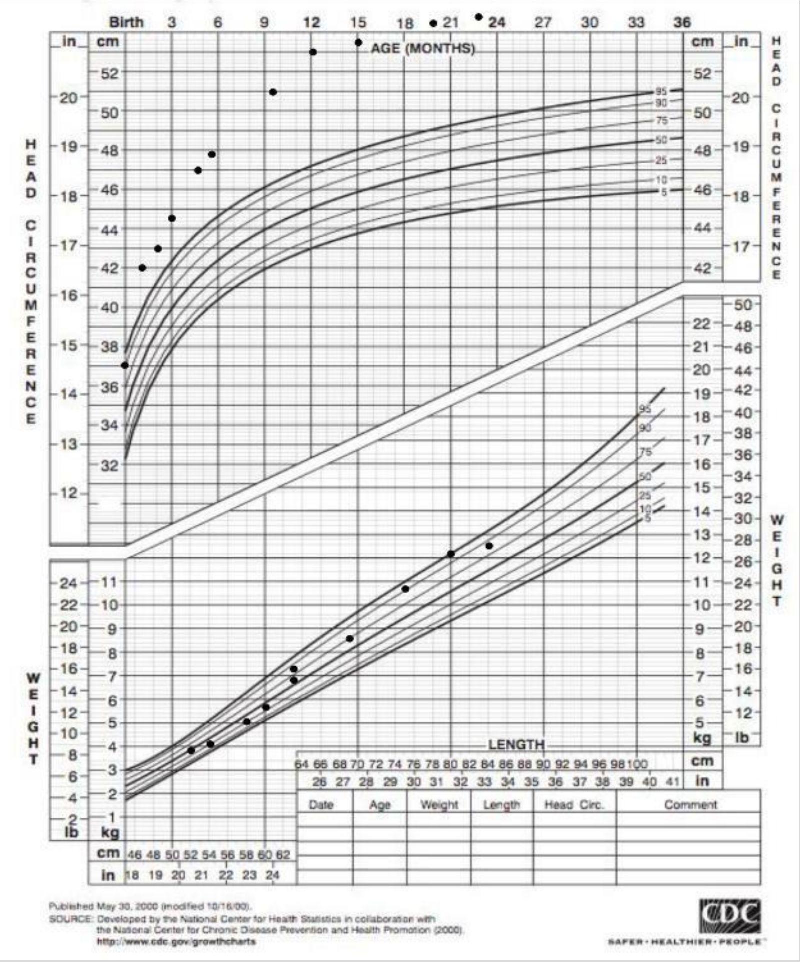
Head Shape and Size

- Microcephaly
  - TORCH
  - Genetic
  - Poor brain growth
- Macrocephaly
  - Benign Familial
    - (measure parents heads)
  - Hydrocephalous
- Abnormal Head Shape



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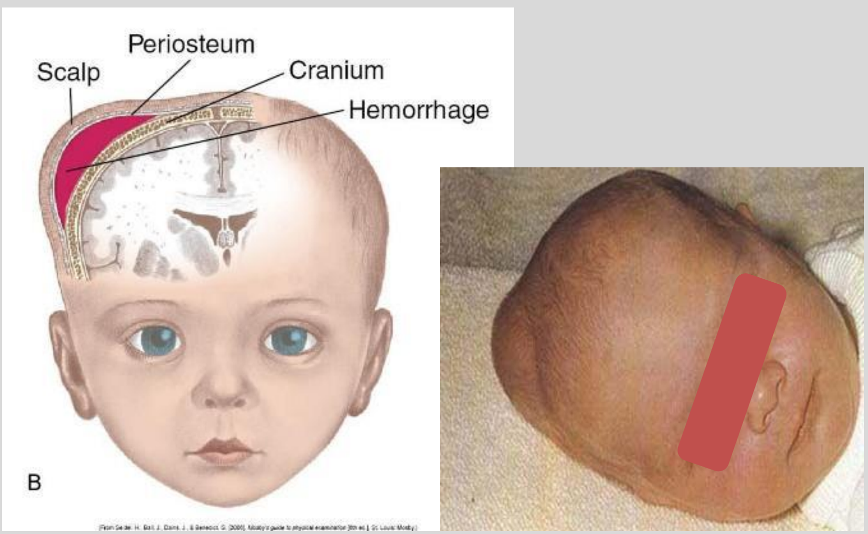
Cephalohematoma

- Doesn't cross suture lines
- Well circumscribed
- Reassure, maybe residual calcification



Caput Succedaneum

- Diffuse edematous swelling
- "Cap" across suture lines
- Observation and reassurance

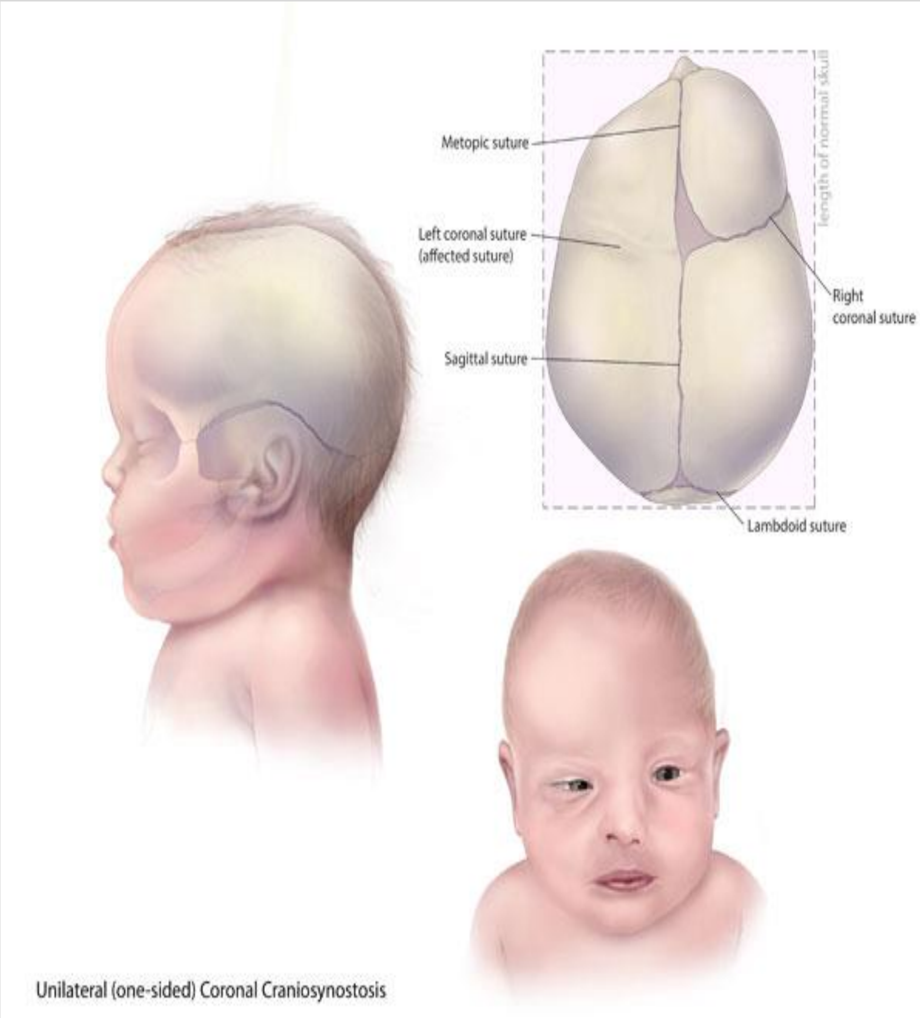
Head Shapes…still cute!

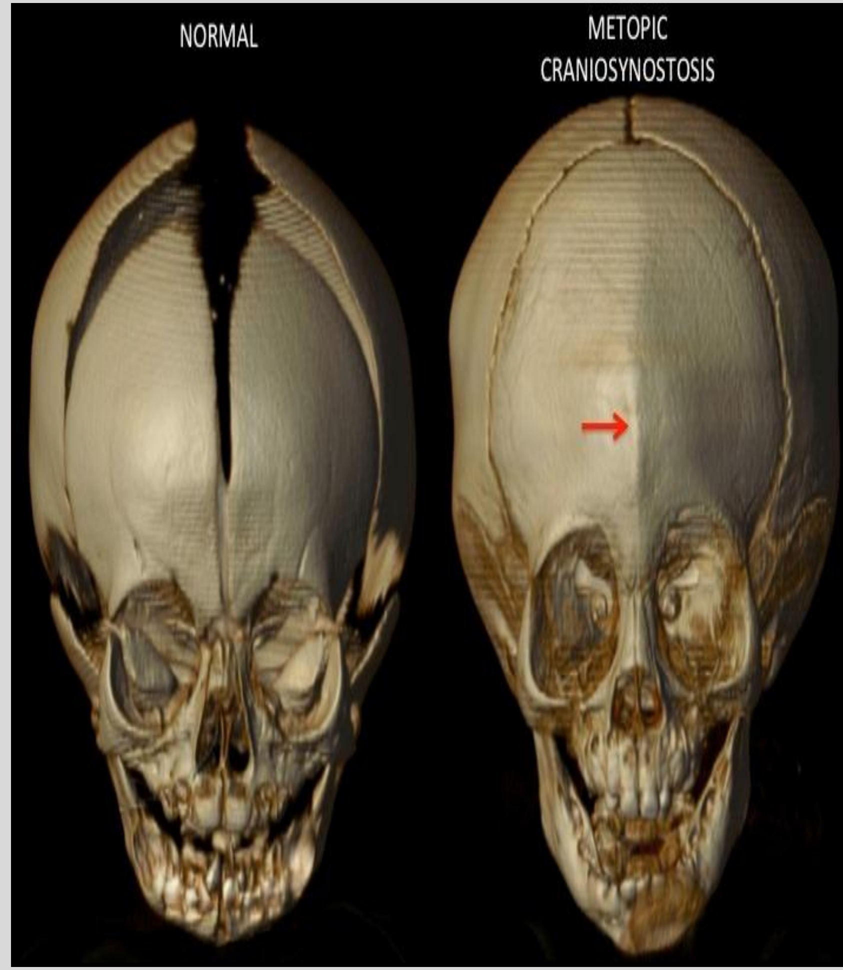
- Micro Premie common **dolichocephaly**
- **Brachycephaly-** short, wide head, can be positional or coronal suture premature fusion
- **Positional plagiocephaly very common**
  - More belly time when awake, ROM neck exercises if torticollis, Helmet sometimes
- Still need to sleep on back for SIDS

Craniosynostosis

- Premature or abnormal fusion of Cranial sutures
- Ridging of sutures
- Asymmetric head and shapes, forehead, ears, eyes, (proptosis), nose
- Get 3D CT scan, neurosurgical consult

Craniosynostosis

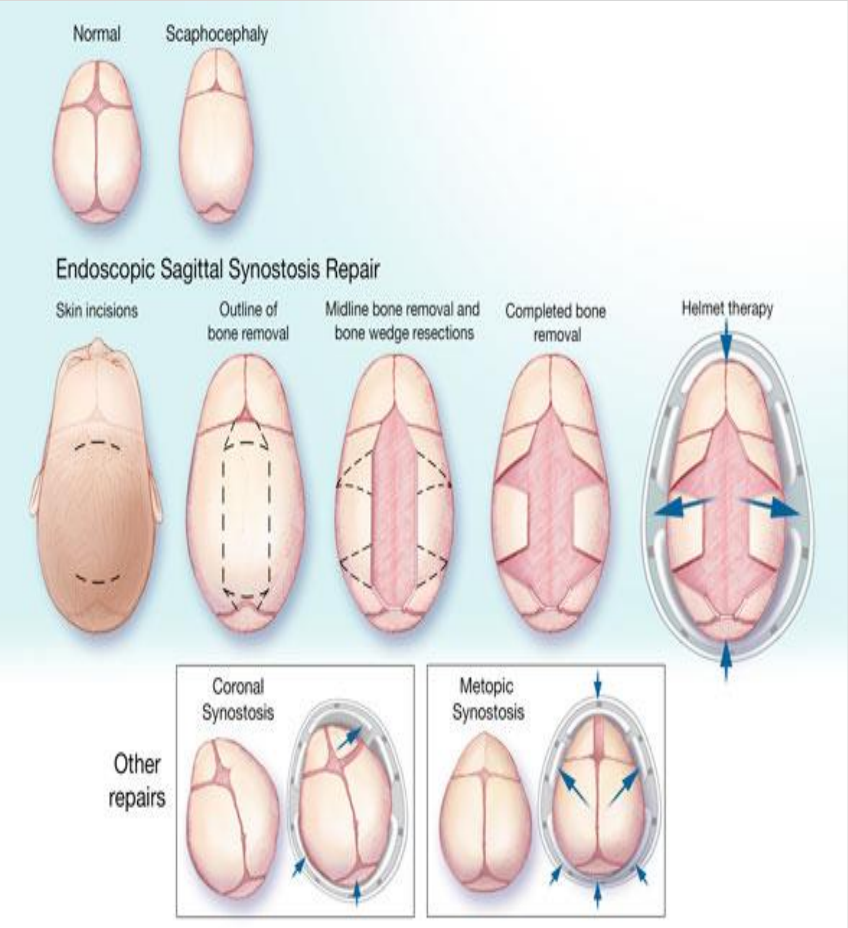




Treatment

Helmets

Surgical



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Craniosynostosis- syndromes

Crouzon

- Asymmetrical face, forehead, ears
- Eyes- Proptosis, hypertelorism

Apert

- Craniosynostosis
- Limbs affected

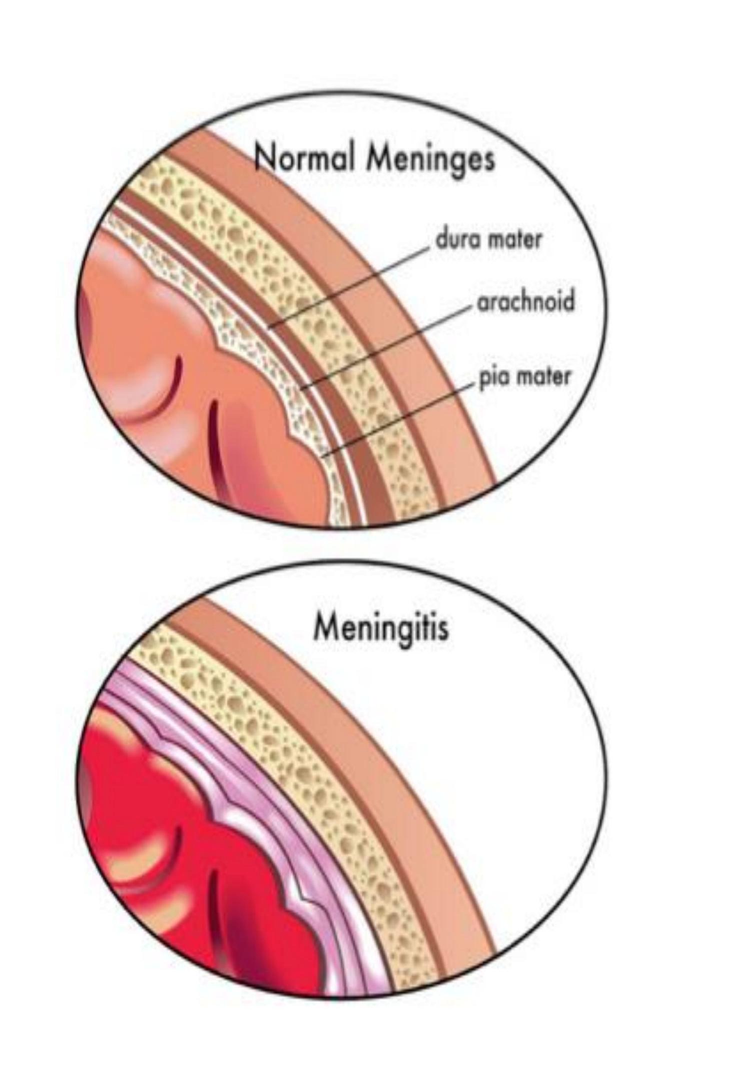
Infections with Neurologic Problems

- TORCH infections
- Meningitis
- Bell Palsy
- Roseola and Shigella (assoc with febrile seizures)

- **T**oxoplasmosis
- **O**ther (Parvo,VZV,TB)
- **R**ubella
- **C**MV
- **H**erpes/**H**IV/**H**epatitis
- Syphillis

Meningitis

- Symptoms: headache, nausea, vomiting, anorexia, restlessness, altered state of consciousness, irritability
- Signs of CNS infection: fever or hypothermia, photophobia, neck pain, rigidity, obtundation, stupor, coma, seizures, focal neurologic defects
- Signs of increased ICP: headache, emesis, **bulging fontanel,** widening of sutures, oculomotor or abducens nerve palsies, **apnea, hyperventilation**
- Cushing's triad (Irregular respirations, Hypertension, Bradycardia)

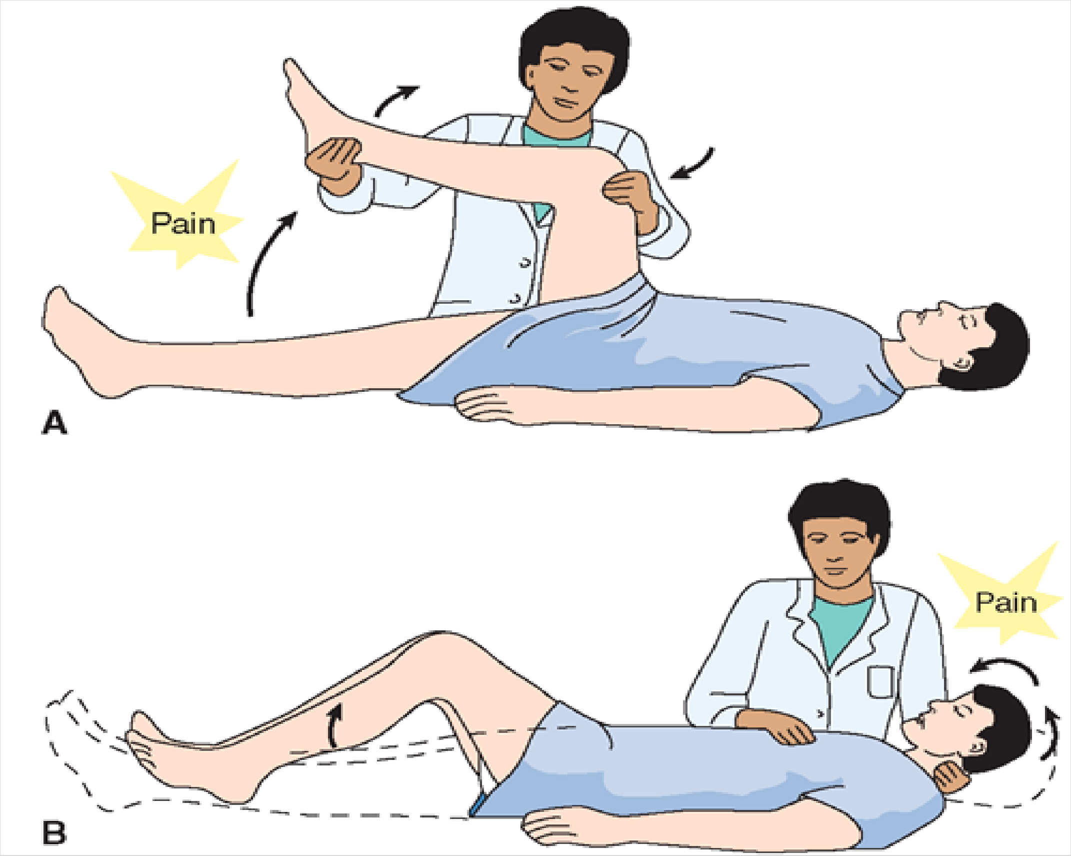


Meningitis in Infants

- Irritable
- Lethargy
- Inconsolable
- Temperature (**fever** or **hypothermic**)
- Vomiting
- Apnea
- Seizures

Meningitis Physical Exam Signs

Not reliable in infants, young children!

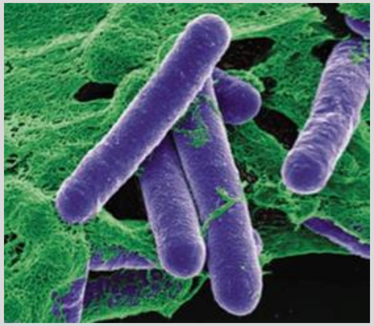


Kernig's sign

Brudzinski's sign

Infantile Botulism

- *Clostridium botulinum*: anaerobic, gram (+), spore forming bacillus
- Occurs in infants younger than 12 months
- Spores in Honey! No honey under 1 yr old!
- Physical findings: constipation, progressive hypotonia, lethargy, feeding problems, extraocular movement impairment, drooling, poor suck and gag reflex, flushed skin, NO fever, possible respiratory failure
- Toxin inhibits acetylcholine release
- Botulism Immune Globulin IV (BIG-IV), supportive care



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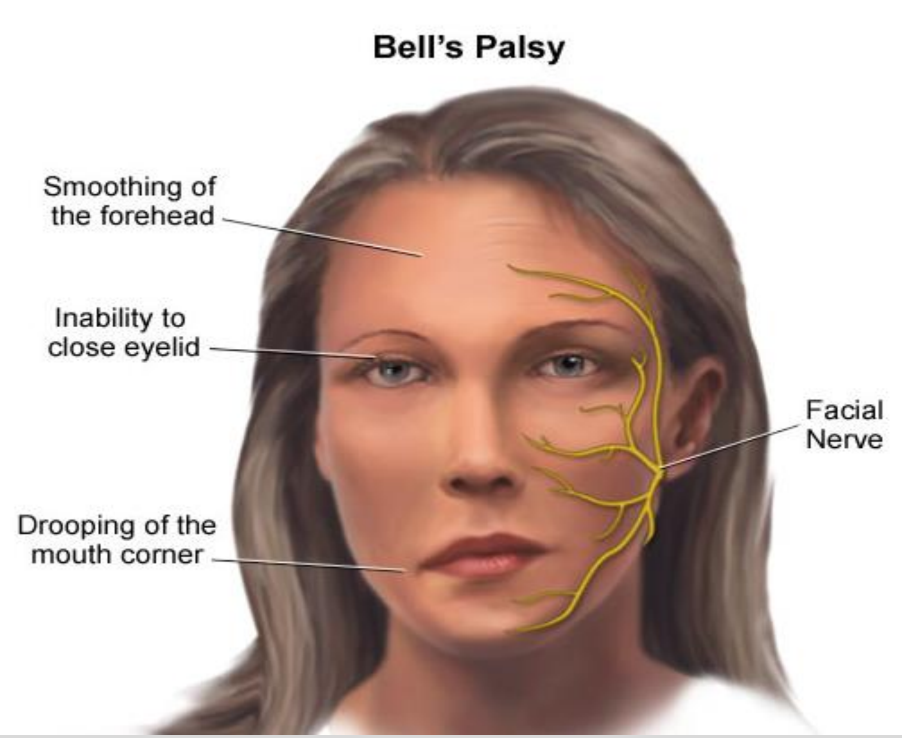
Bell Palsy

- Unilateral facial nerve palsy
- Often develops around 2 weeks after a systemic infection (**herpes simplex**, **varicella-zoster**, Epstein-Barr, **Lyme**, mumps, *Mycoplasma, others*)
- Treatment- **eye drops/ patch**, oral prednisone



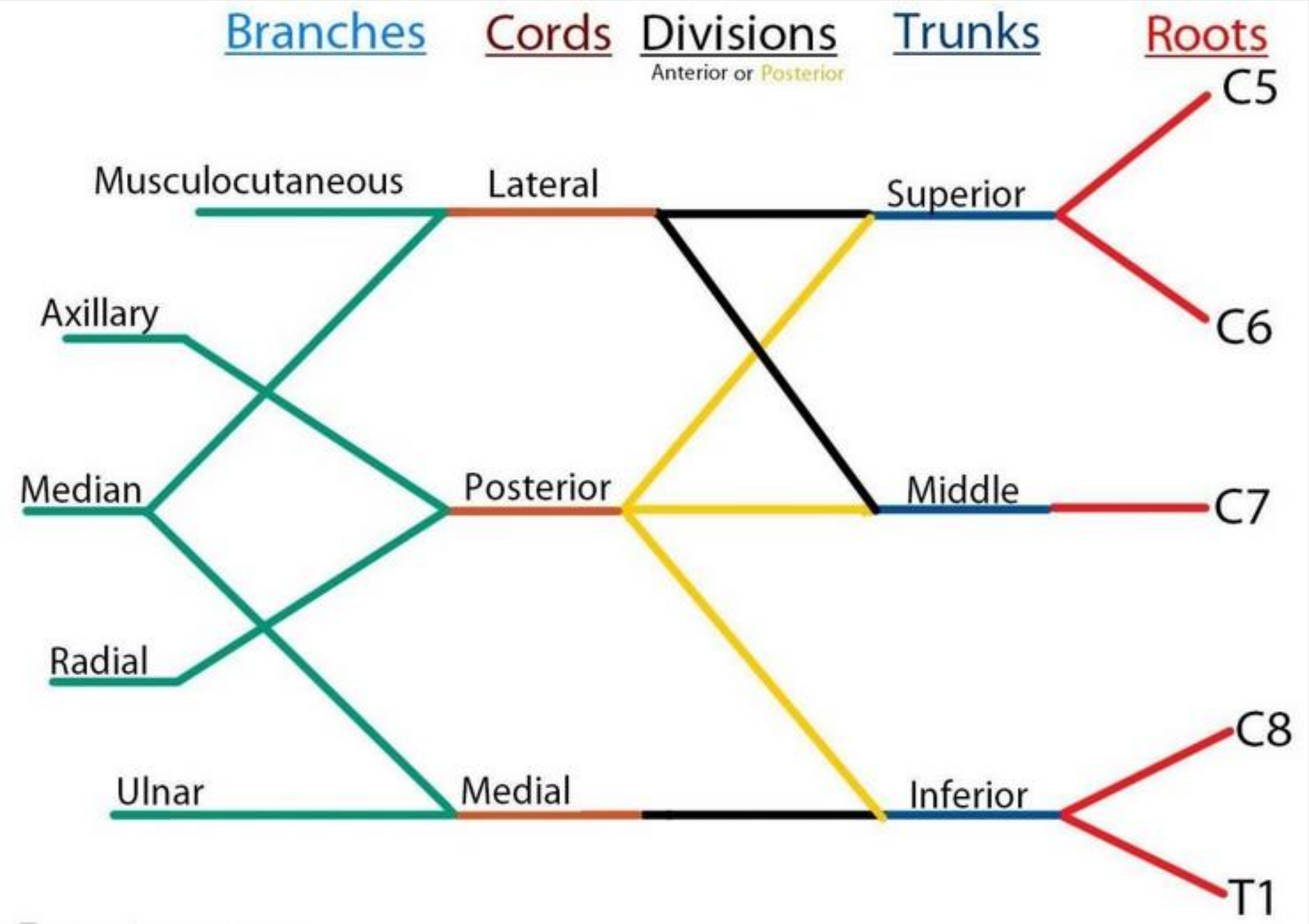
Asymmetric smile when crying -otherwise well

- Congenital Absence of the Depressor angularis oris muscle
  - reassurance, gets less noticeable over time
  - minor association with Cardiac anomalies

Trauma/ Injury

- Erb's Palsy
- Klumpke's Palsy
- Cephalohematoma
- Caput Succedaneum
- Facial nerve palsy (forceps)
- Shaken Baby Syndrome
- Concussions

Brachial Plexus



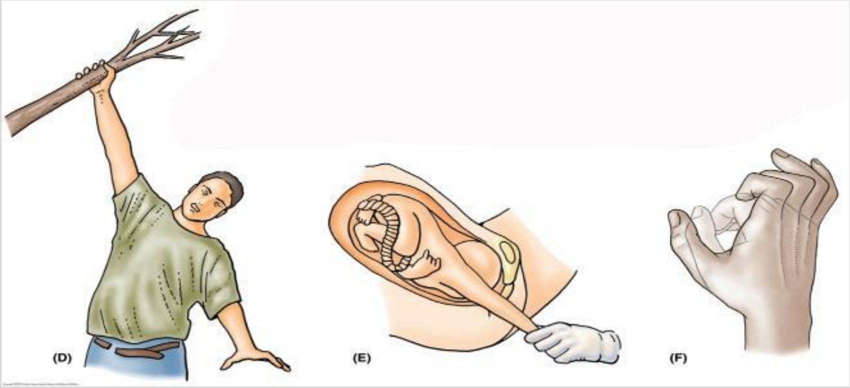
Erb's Palsy

- Injury to upper trunk (**C5-6)-** damage to musculocutaneous, axillary, and suprascapular nerves
- **Arm hangs by side** (paralysis of abductors), **internally rotated** (paralysis of external rotators), **with forearm pronated** (loss of biceps)
- Hand function preserved(nl grip)
- Often complication of difficult birth
- **Asymmetric Moro reflex!**

- Waiter's tip

Klumpke's Palsy- Worse prognosis

- Injury to lower trunk (C8-T1)- damage to ulnar nerve
- **Weakness of intrinsic hand muscles** (thenar, hypothenar, lumbricals, interossei)
- Claw hand deformity
- Caused by: hyperabduction injuries, cervical rib (thoracic outlet syndrome)
- +/- Horner's syndrome (miosis, ptosis, anhidrosis)



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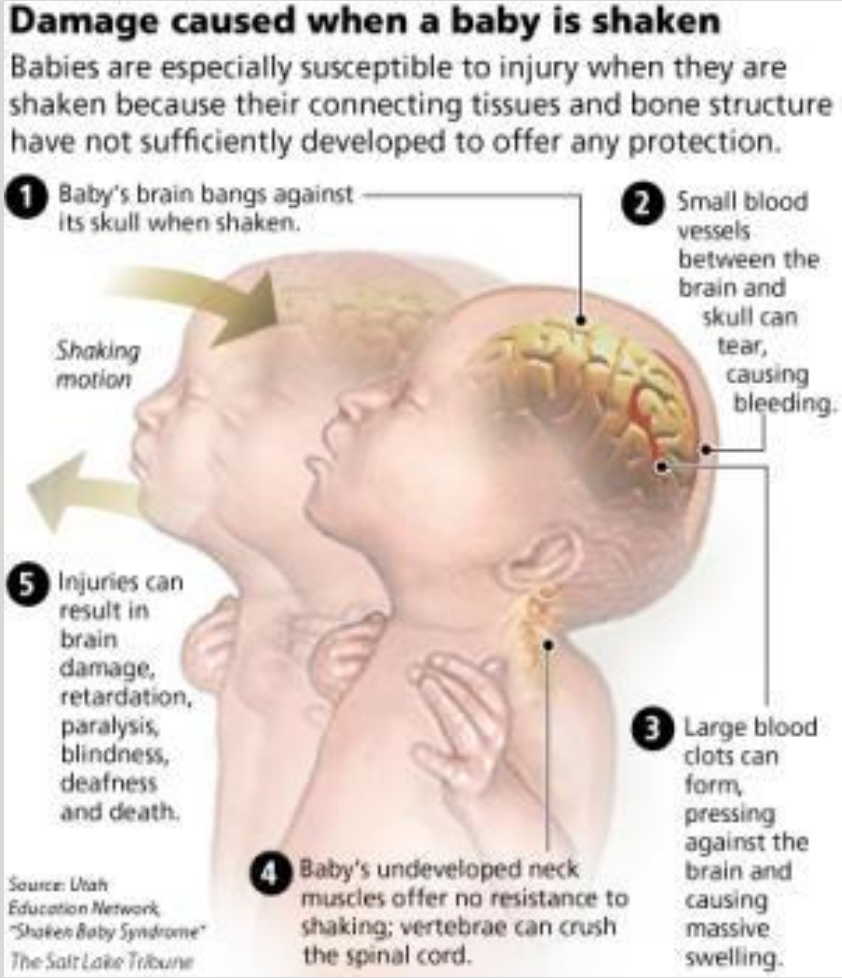
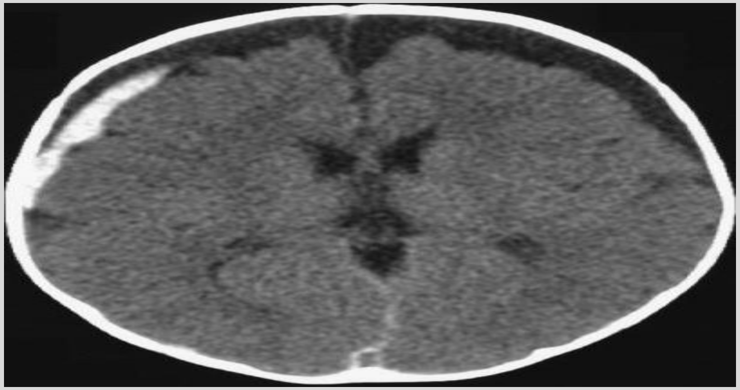
Shaken Baby Syndrome

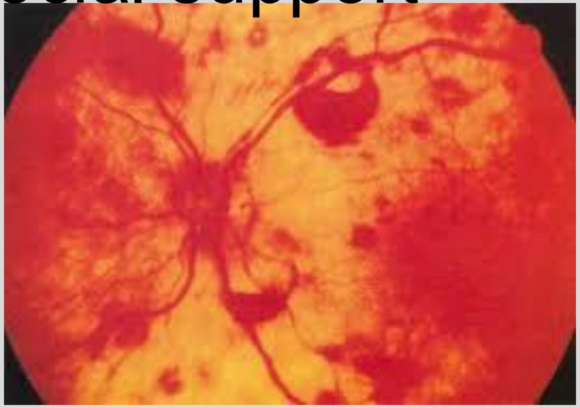
- Intracranial injury caused by rapid acceleration and rotation of cranium
- Shearing injury of brain/intracranial vessels
- Symptoms: drowsiness, lethargy, decreased feeding, vomiting, apnea, seizures
- Physical manifestations: **subdural hematoma, retinal hemorrhage**



Shaken Baby Syndrome

- Part of Anticipatory guidance
- Watch for cranky/colic infants, cranky adults, poor coping mechanisms, lack of social support



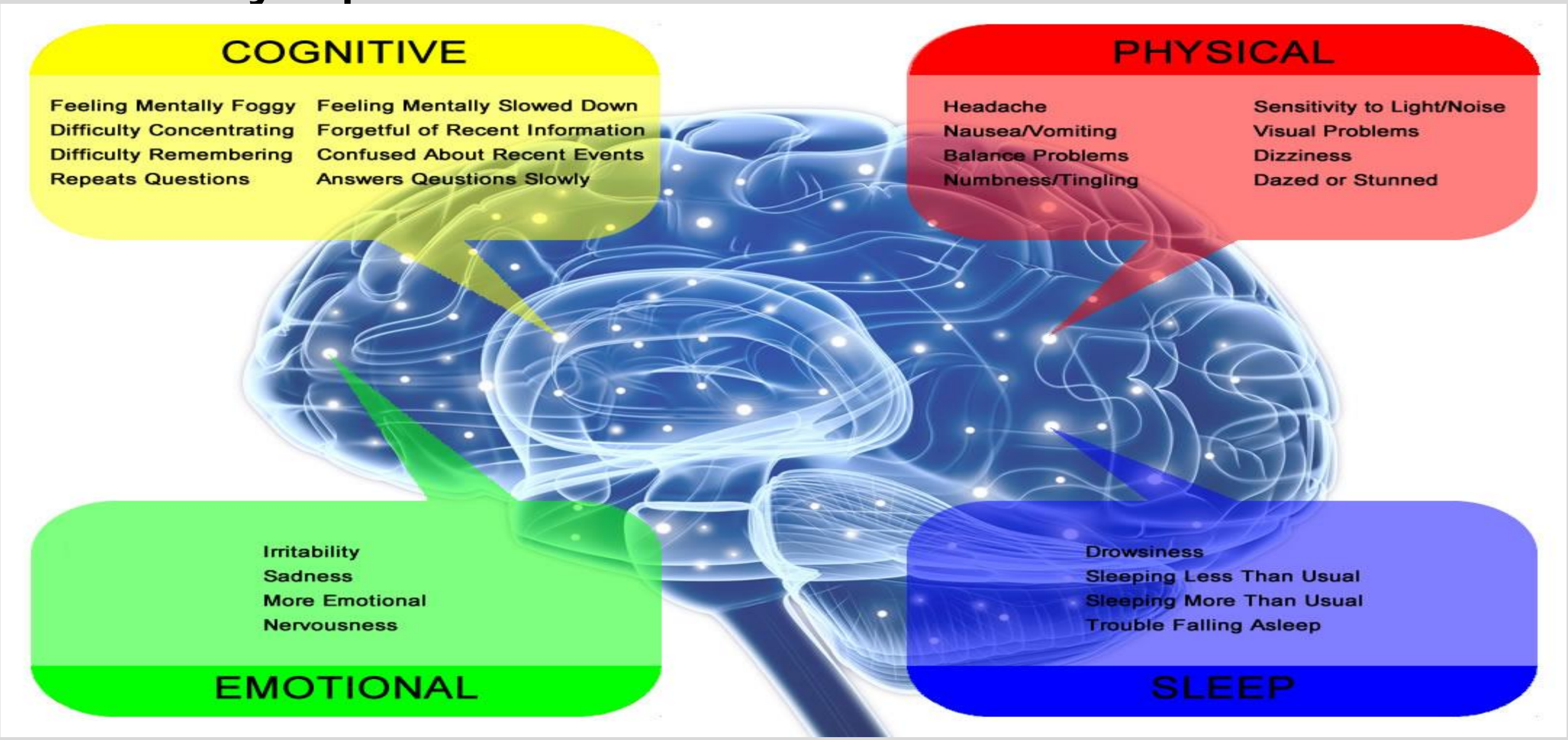


Concussion

- Traumatic forces on brain result in short-lived neurologic impairment that typically resolves spontaneously
- Immediate or evolves over minutes to hours
- Symptoms can include somatic, cognitive, emotional, physical, and sleep disturbance. Photosensitivity.
- Treatment:
  - physical rest- no work or school, no activities early on
  - cognitive rest- no reading, no Texting/TV, dark room, sunglasses, no loud music/noise, quality sleep/rest
  - Can engage in some light low intensity activities if feeling better (Prolonged rest may prolong symptoms.). Build back slowly.
  - recovery time varies.

Concussions

- When can they return to sports???
  - When they are 100% better and asymptomatic with a normal Neuro exam



Summary Slide 1

- Tests for hypotonia- Prone, supine, hold infant up- if slips thru fingers, then hypotonic
- Cerebral Palsy- non progressive motor impairment, upper motor neuron signs- spasticity, increased DTR's, clonus
- Spinal Muscular Atrophy (SMA) anterior horn cell, severe hypotonia, normal intelligence, fasciculations of tongue, poor feeding
- Down Syndrome- Hypotonia and other physical signs
- Prader Willi- Paternal 15q11.2-q13), infant hypotonic and then won't stop eating later in life, dev delays
- Dermatomyositis- Classic skin findings with proximal muscle weakness
- Duchenne MD- Problem with dystrophin, Weakness develops in early childhood, males, pseudohypertrophy in claves, Elevated CK, cardiomyopathy and pulmonary issues
- Becker- milder form of MD

Summary Slide Part 2

- Cephalohematoma doesn’t cross suture lines,
- Caput Succedaneum does cross
- Positional plagiocephaly More belly time when awake, ROM neck exercises if torticollis, Helmet sometimes, still sleep on back
- Craniosynostosis- asymmetry to face and head, ridging of sutures, Get 3D CT scan and neurosx consult
- Meningitis- bulging fontanel, fever or hypothermia, apnea, lethargy/irritable
- Botulism- No honey under 1 yr old, hypotonia, difficulty eating, drooling, poor suck or gag reflex
- Bells palsy- protect the eye from drying out
- Erb’s palsy- asymmetric moro reflex
- Klumpke palsy- loss of grip, can have a horner syndrome
- Shaken baby- subdural hematomas, retinal hemorrhages
- Concussion- Varied symptoms, need full recovery to go back to full activity

Lecture Feedback Form:

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THE END!